

An Item-Level Audit Protocol for Error Persistence and Regression Burden in Small Language Model Replacement

DHANANJAY SINGH CHAUHAN*, Independent Reviewer

Replacement decisions for small language models are often justified by aggregate benchmark gains. That practice can hide a deployment-relevant distinction: a candidate model may improve the mean score while newly failing on items that the current model answered correctly. This software-tool article presents a paired item-level replacement-audit protocol. For each shared labeled item, the protocol separates persistent correct answers, corrected errors, regressions, and persistent errors, then reports correction rate, regression mass, churn mass, error persistence, and normalized regression burden.

The contribution is a practical reporting protocol and public artifact, not a scaling law or a causal account of model families. The primary evidence is a parser-independent WILD correctness-only use case with 276,685 item-level records for five Qwen and Llama models up to approximately four billion parameters across ten tasks. In this public setting, Qwen2.5-0.5B to Qwen2.5-3B improves aggregate correctness by 22.2 percentage points while changing 40.9% of item outcomes; Llama 3.2 1B to 3B improves by 17.8 points while changing 35.3% of outcomes; and the near-parity Qwen2.5-1.5B to Qwen2.5-3B comparison improves by 5.7 points while changing 30.4% of outcomes. A secondary archived-output MMLU-Pro use case reconstructs 3,008 source items across 14 model-output runs and demonstrates raw-response audit tooling, but remains diagnostic because complete inference manifests and human parser validation are not yet available. The core claim is narrow: benchmark-based replacement decisions should report item-level corrections, regressions, persistent errors, and churn alongside aggregate accuracy.

1 Introduction

Small language models are attractive deployment targets because they can satisfy memory, latency, privacy, and cost constraints that make larger models impractical. In that setting, replacement is an ordinary engineering decision. A team has a current model, evaluates a candidate model, and must decide whether the candidate is a better component for the same task. Benchmark accuracy is often the first summary used for this decision.

Aggregate accuracy is a useful summary, but it is not a replacement audit. The same net gain can arise from very different item-level behavior. A candidate may correct many current-model errors while introducing few new failures, or it may achieve a similar mean score after replacing one set of correct items with another. The second case matters in applications where previously reliable cases carry operational risk, such as routing documents, flagging compliance cases, or triaging support requests. A higher average score does not tell a user which old errors remain, which old errors were fixed, or which old successes were lost.

This paper treats model replacement as a paired item-level accounting problem. Given two models evaluated on the same labeled item identifiers, each item falls into one of four cells: persistent correct, corrected error, regression, or persistent error. From these counts, the audit reports aggregate gain together with improvement mass, regression mass, churn mass, error persistence, correction rate, and normalized regression burden. The result is not a new training method or a mechanistic explanation of model behavior. It is a reporting protocol for a practical question: what changes, item by item, when one small model is used in place of another?

The empirical design follows that modest claim. The primary demonstration uses WILD, a public item-level correctness matrix, because binary correctness records avoid the answer-extraction problem that arises with raw language-model completions. The retained WILD subset contains five Qwen and Llama models up to approximately four billion parameters across ten tasks. It is not presented as a representative census of all small models. It is used to show, in a public parser-independent setting, that aggregate gains and item-level churn are distinct quantities.

*Corresponding author.

The paper also includes an archived-output MMLU-Pro case study. That case study is useful because it shows the additional work required when the audit begins from raw model responses: source-row reconstruction, answer extraction, scoring-mode sensitivity, parser-audit sampling, and run-provenance checks. It is not treated as confirmatory evidence for parser-dependent claims, because the original run logs do not contain complete inference manifests and the manual parser audit is not complete. This evidence split is intentional. WILD carries the core item-level correctness claim; MMLU-Pro demonstrates the raw-output workflow and its remaining validation requirements.

The central question is therefore deliberately narrow:

When a candidate small language model improves a benchmark score, how much of the shared item-level correctness pattern is corrected, regressed, or left unchanged?

The paper makes four contributions:

- (1) A reusable replacement-audit protocol for paired item-level correctness, including correction rate, regression mass, churn mass, error persistence, and normalized regression burden.
- (2) A clear distinction between deployment-path audits and standardized all-pairs diagnostics, so directional metrics are interpreted only with respect to a specified current model and candidate model.
- (3) A public WILD correctness-only demonstration showing that aggregate improvements can coincide with substantial item-level churn without relying on private completions or answer parsing.
- (4) A diagnostic archived-output MMLU-Pro case study with source-item hashes, scoring-mode sensitivity checks, parser-audit tooling, and explicit limits on answer extraction and run provenance.

This submission is framed as a software-tool article: the main object is the audit protocol and public artifact, with WILD serving as the primary public use case and MMLU-Pro serving as a diagnostic raw-output workflow.

The paper does not claim that prediction churn is new, that one model family has a general property, or that the reported values define a scaling law. Its claim is operational: when benchmark scores are used to justify replacing a model, the report should include corrected errors, regressions, persistent errors, and churn, not aggregate accuracy alone.

2 Related Work

Beyond aggregate accuracy. NLP evaluation has repeatedly shown that aggregate scores can hide important behavior. CheckList introduced behavioral tests to expose failures hidden by held-out accuracy (Ribeiro et al. 2020). HELM argues for transparent, multi-metric model evaluation rather than single-number comparisons (Liang et al. 2022). The present audit follows the same evaluation philosophy, but focuses on paired item-level changes between a current model and a candidate replacement model.

Prediction churn and regression bugs. Prediction churn studies how model predictions change across retraining or deployment iterations (Fard et al. 2016). Regression-bug work in NLP model updates measures negative flips: cases where a newer model newly fails examples an older model handled correctly (Xie et al. 2021). Unlike work that proposes training or update methods to reduce regressions, this paper proposes a reporting and audit artifact for replacement decisions. The present work does not claim to invent churn or regression measurement. Its narrower contribution is to adapt this accounting to small language model replacement and to keep beneficial churn, harmful churn, and persistent old errors separate in the artifact and reporting.

Regression testing and deployment audit. The audit is close in spirit to software regression testing. Before replacing a component, users need to know which previously passing cases now fail. In model evaluation, the analogous question is which items were previously correct and are now wrong. This is a deployment-facing question rather than a claim about training mechanisms.

Tier	Contents	Permitted use
A	Raw prompts, raw completions, labels, item IDs, and validated scoring	Full audit, parser analysis, format-failure analysis
B	Item IDs, model IDs, and binary correctness	Correctness-only churn, regression, persistence, and correction metrics
C	Aggregate benchmark scores only	Context only; no item-level replacement claims
D	Gated, unavailable, incompatible, or unverified data	Excluded until access and terms are clear

Table 1. Evidence tiers for replacement accounting. Only Tier A and Tier B sources support item-level churn metrics.

Production readiness and documentation. Production ML work emphasizes systematic testing, monitoring, and documentation rather than isolated aggregate scores (Breck et al. 2017; Sculley et al. 2015). Dataset and model documentation frameworks likewise stress provenance, intended use, and evaluation context (Geburu et al. 2021; Mitchell et al. 2019). The present artifact follows that orientation by separating source provenance, evidence tier, scoring status, and claim use.

Stability, forgetting, and underspecification. The replacement problem also connects to work on catastrophic forgetting (Kirkpatrick et al. 2017) and underspecification (D’Amour et al. 2022). Those literatures study different mechanisms and failure modes. Here, the object is narrower: observable item-level correctness changes between archived model outputs or public correctness records.

Benchmarks and public item-level resources. Benchmark suites such as MMLU, MMLU-Pro, and BIG-bench provide broad task-level measurement (Hendrycks et al. 2021; Srivastava et al. 2022; Wang et al. 2024). WILD provides public item-level correctness records that can be used for correctness-only replacement audits (Krumdick et al. 2026). Aggregate leaderboards and model cards are useful for context, but they cannot identify which old errors were corrected or which previously correct items regressed.

3 Replacement-Audit Protocol

3.1 Inputs and evidence tiers

The audit requires paired item-level outcomes for two models on the same labeled item identifiers. The strongest input is raw model output plus a validated scoring procedure, meaning either benchmark-provided predictions or a completed human-labeled parser audit. A weaker but still useful input is a public binary correctness matrix. Aggregate leaderboards alone are insufficient.

3.2 Four-cell decomposition

For a directed comparison from current model A to candidate model B over the same N items, each item belongs to exactly one cell:

- persistent correct (PC): both models are correct;
- improvement or corrected error (I): A is wrong and B is correct;
- regression (R): A is correct and B is wrong;
- persistent error (PE): both models are wrong.

The aggregate accuracy delta is

$$\Delta = \frac{|I| - |R|}{N}. \quad (1)$$

Evidence component	Status	Claim use
WILD item-level correctness	Supported	Core replacement-accounting evidence for persistence, regression mass, and churn over paired correctness records.
MMLU-Pro source reconstruction	Supported	Provenance evidence that all 3,008 evaluated item identifiers map to the stated MMLU-Pro snapshot.
MMLU-Pro raw-output accounting	Diagnostic only	Demonstrates the raw-output workflow and artifact-risk checks, but remains parser-dependent.
Parser-validation claims	Not claimed	Human labels are incomplete, so the paper does not report parser agreement as validation.
Aggregate benchmark context	Context only	Positions model capability; does not support item-level replacement or churn claims.

Table 2. Claim ledger for the current GitHub/SSRN release. Parser-dependent MMLU-Pro claims are explicitly excluded from the core claim set until human validation is complete.

The audit metrics are

$$\text{Improvement mass} = \frac{|I|}{N}, \quad (2)$$

$$\text{Regression mass} = \frac{|R|}{N}, \quad (3)$$

$$\text{Churn mass} = \frac{|I| + |R|}{N}, \quad (4)$$

$$\text{Error persistence} = \frac{|PE|}{|PE| + |I|}, \quad (5)$$

$$\text{Correction rate} = \frac{|I|}{|PE| + |I|}. \quad (6)$$

Let the current model’s error rate be

$$e_0 = \frac{|PE| + |I|}{N}. \quad (7)$$

The normalized regression burden is

$$\text{NRB} = \frac{|R|/N}{e_0}. \quad (8)$$

If $|PE| + |I| = 0$, the current model has no observed errors on the shared item set. Error persistence, correction rate, and NRB are then undefined and are reported as NA, not as zero. Category summaries with zero current-model errors remain included for mass metrics such as improvement mass, regression mass, and churn mass, but old-error-normalized ratios are reported as NA.

For an accuracy-improving directed comparison, NRB is a practical normalization of regression mass by the current model’s observed error rate. It can also be read descriptively as the gap between the no-regression upper bound on old-error persistence and observed old-error persistence. This interpretation is descriptive; it is not a theoretical novelty claim and does not identify a causal mechanism.

3.3 Directed paths versus all-pairs diagnostics

The metrics are directional. A comparison $A \rightarrow B$ answers a different deployment question than $B \rightarrow A$. A deployment audit should therefore use the actual proposed replacement path whenever it is known.

Field	Purpose
Current and candidate model accuracy	Aggregate movement
Accuracy delta	Net gain or loss
Improvement mass	Share of all evaluated items that the current model missed and the candidate model corrected
Regression mass	Share of all evaluated items that the current model answered correctly and the candidate model newly misses
Churn mass	Total share of changed item outcomes
Error persistence and correction rate	How much of the current error set remains
Normalized regression burden	Practical normalization of regression mass relative to current-model errors
Top regressing categories or clusters	Where manual review should start
Path note	Whether the directed pair is a real replacement path or a diagnostic comparison

Table 3. Minimal replacement-audit card emitted by the artifact.

When only a set of model outputs is available, a standardized all-pairs diagnostic can still be useful. In that case, the ordering rule must be explicit and should not be mistaken for vendor release chronology. In the MMLU-Pro case study below, directed pairs are ordered by parameter count, then generation string, then model label. Comparisons marked as chronology unclear are treated as diagnostic pairs, not as true deployment histories.

3.4 Minimal audit-card algorithm

- (1) Align current-model and candidate-model records by item identifier.
- (2) Convert each model’s output to a binary correctness label under a documented scoring procedure.
- (3) Count PC, I, R, and PE.
- (4) Compute Δ , improvement mass, regression mass, churn mass, error persistence, correction rate, and NRB.
- (5) Emit an audit card with optional review flags for high churn, high regression mass, high NRB, near-parity churn, and negative-delta swaps.

3.5 Statistical posture

The audit is primarily descriptive. McNemar-style paired tests are appropriate when comparing two classifiers on the same items (Dietterich 1998; McNemar 1947), but replacement accounting should not treat shared-endpoint model pairs as independent evidence. The artifact therefore reports mass and churn metrics as the primary outputs and uses inferential checks only as clearly labeled diagnostics.

4 Implementation and Operation

The public artifact is structured as a small analysis package rather than a monolithic notebook. Each command writes CSV, JSON, and paper-table outputs that can be inspected independently. Table 4 summarizes the main modules, their input requirements, and their role in the paper.

The public package targets Python 3.11 or newer and uses ordinary scientific Python dependencies listed in requirements.txt. From the release root, the shortest public workflow is:

```
make reproduce-public
make test
make hygiene
```

Module	Input	Output	Paper use
WILD ingest	WILD item-level correctness records	Normalized records, pairwise replacement metrics, task-dispersion tables	Core
Evidence registry	Source registry entries	Evidence-tier map and claim-use table	Claim boundary
MMLU-Pro manifest	Derived MMLU-Pro item files and source parquet	Redacted source-row hashes and manifest status	Provenance
Paper metrics	Derived MMLU-Pro correctness summaries	Pairwise metrics, audit cards, appendix tables	Diagnostic
Parser-impact audit	Private parser-audit labels, when available	Parser-impact report and claim gate	Diagnostic gate
Release hygiene	Public release directory	Leakage scan for raw responses, benchmark text, private samples, and local paths	Release safety

Table 4. Software modules used by the public artifact. The WILD module validates the pairwise accounting layer; raw-output parser validation is a separate diagnostic layer and is not claimed complete.

Field	Example output
Current model	Qwen2.5-1.5B-Instruct
Candidate model	Qwen2.5-3B-Instruct
Shared WILD items	54,464
Accuracy delta	+5.7 percentage points
Improvement mass	18.0%
Regression mass	12.3%
Churn mass	30.4%
Interpretation	Near-parity aggregate gain with substantial item-level churn

Table 5. Worked WILD audit-card example for a selected-task pooled same-family comparison. The example illustrates how a small aggregate gain can still require reviewing a large changed item set.

The first command regenerates the public WILD and source-registry outputs, the second runs the test suite, and the third verifies that the public release does not contain raw model responses, benchmark text, private parser-audit samples, or local cache paths. The package also includes a `manifest` target that refreshes file hashes for the public release.

5 Use Case 1: WILD Correctness-Only Audit

The WILD demonstration is the parser-independent empirical support for the audit protocol because it avoids answer extraction. WILD exposes item identifiers, model identifiers, task identifiers, and binary correctness records. The full WILD release contains evaluations of 65 models over 109,564 unique items spanning 163 tasks from 27 datasets (Krumdick et al. 2026). The present artifact uses a selected public subset containing five Qwen and Llama models up to approximately four billion parameters and ten tasks. It supports item-level churn accounting, but it does not support raw-response parsing claims, prompt-format claims, or manual answer-extraction validation.

Model	Family	Size	Tasks	Rows	Included	Reason
Qwen1.5-4B-Chat	Qwen	4.0B	10	55337	no	Different Qwen generation; not part of the artifact-defined Qwen2.5 replacement chain.
Qwen2-1.5B-Instruct	Qwen	1.5B	10	55337	no	Single Qwen2 small model in WILD; no retained same-generation small replacement pair.
Qwen2.5-0.5B-Instruct	Qwen	0.5B	10	55337	yes	Retained in the artifact-defined Qwen2.5 size chain.
Qwen2.5-1.5B-Instruct	Qwen	1.5B	10	55337	yes	Retained in the artifact-defined Qwen2.5 size chain.
Qwen2.5-3B-Instruct	Qwen	3.0B	10	55337	yes	Retained in the artifact-defined Qwen2.5 size chain.
granite-3.2-2b-instruct	Granite	2.0B	10	55337	no	No second under-4B Granite model in WILD for a same-family replacement pair.
llama-3.2-1b	Llama	1.0B	10	55337	yes	Retained in the Llama 3.2 small-model replacement pair.
llama-3.2-3b	Llama	3.0B	10	55337	yes	Retained in the Llama 3.2 small-model replacement pair.
Nemotron-Mini-4B-Instruct	Nemotron	4.0B	10	55337	no	No second under-4B Nemotron model in WILD for a same-family replacement pair.

Table 6. WILD inclusion coverage for public models at approximately four billion parameters or below. The table reports selected-task rows available under the artifact extraction rule and the reason each model was retained or excluded from the paired-family demonstration. The complete machine-generated inclusion and exclusion files are included in the public release.

Field	Value
External source	WILD
Evidence tier	B: item-level correctness
Models retained	5
Tasks retained	10
Normalized records	276685
Pairwise comparisons	44 (4 selected-task pooled + 40 task-level)
Selected-task pooled Qwen mean churn	34.4%
WILD source scale	65 models; 109,564 items; 163 tasks; 27 datasets

Table 7. Public WILD correctness-only demonstration.

The WILD use case validates the pairwise accounting layer of the artifact, not the raw-output scoring layer. The subset rule is fixed in the artifact: retain public item-level correctness rows for same-family small-model chains with at least two public models at or below approximately four billion parameters, shared coverage over the selected tasks, and a plausible size or generation replacement path. The selected tasks are MMLU-Pro, MMLU, ARC-Challenge, GSM8K, HellaSwag, PIQA, BoolQ, IFEval, BBH, and MATH. Models without a retained same-family comparator are excluded from pairwise analysis but listed in Table 6. This is a public cross-benchmark demonstration of the audit protocol, not a representative analysis of all WILD models or all WILD tasks. No estimate in this section is interpreted as the average churn rate of all WILD models.

Rows in Table 6 are retained model-task correctness records before pairwise intersection; pairwise N in Table 8 is the number of shared item identifiers after aligning each directed model pair.

The generated WILD replication contains 44 directed same-family comparisons: four selected-task pooled comparisons, plus 40 task-level comparisons across ten tasks. Across the selected WILD tasks, the three Qwen selected-task pooled directed comparisons have 34.4% mean churn. The Qwen2.5-0.5B to Qwen2.5-3B comparison improves aggregate correctness by 22.2 percentage points while changing 40.9% of item outcomes and regressing 9.3% of all shared items. The near-parity Qwen2.5-1.5B to Qwen2.5-3B comparison improves aggregate correctness

Family	Current	Candidate	N	Δ	I	R	Churn	Persist.	Corr.	NRB
llama	llama-3.2-1b	llama-3.2-3b	54463	+17.8%	26.5%	8.8%	35.3%	58.1%	41.9%	13.9%
qwen	Qwen2.5-0.5B-Instruct	Qwen2.5-1.5B-Instruct	54464	+16.5%	24.2%	7.7%	31.9%	64.2%	35.8%	11.4%
qwen	Qwen2.5-0.5B-Instruct	Qwen2.5-3B-Instruct	54464	+22.2%	31.5%	9.3%	40.9%	53.3%	46.7%	13.8%
qwen	Qwen2.5-1.5B-Instruct	Qwen2.5-3B-Instruct	54464	+5.7%	18.0%	12.3%	30.4%	64.7%	35.3%	24.2%

Table 8. WILD pooled replacement audits. The table reports selected-task pooled directed comparisons for retained Qwen and Llama model chains. Values are computed from binary correctness records only and do not evaluate raw-response parsing. I is improvement mass, R is regression mass, and NRB is normalized regression burden.

Scope	Tasks	Pairs	Median N	Median churn	Churn IQR	Median R	Median persist.	Highest-churn task
selected-task pooled	10	40	4135	33.9%	[30.0%, 39.4%]	7.4%	47.9%	piqa (41.8%)
llama	10	10	4135	34.8%	[32.8%, 37.8%]	6.9%	45.6%	hellaswag (43.7%)
qwen	10	30	4135	32.6%	[28.0%, 39.8%]	7.4%	50.2%	piqa (42.9%)

Table 9. Task-level dispersion for WILD directed comparisons. The table summarizes the 40 task-level comparisons, separate from the four selected-task pooled comparisons in Table 8, and reports shared-item denominators.

by 5.7 percentage points while changing 30.4% of item outcomes. This is the core operational result in a parser-free setting: aggregate gain and item-level replacement churn are different quantities.

The Llama 3.2 1B to 3B comparison gives the same qualitative lesson. It improves by 17.8 percentage points while changing 35.3% of item outcomes and regressing 8.8% of all shared items. The audit therefore identifies not only whether a candidate model wins on average, but how much of the correctness pattern is rewritten.

The task-level dispersion table shows why an all-task mean is not enough. Across all selected WILD task-level comparisons, median churn is 33.9% with an interquartile range of 30.0–39.4%. For Qwen alone, median task-level churn is 32.6%, and the highest mean-churn task among the selected tasks is PIQA at 42.9%. These values do not imply a universal churn level for all small models. They show that the accounting method can be applied to a public item-level correctness matrix and that large churn appears outside the MMLU-Pro archived-output case study.

6 Use Case 2: MMLU-Pro Raw-Output Diagnostic

The MMLU-Pro case study uses 14 archived model-output runs over 3,008 shared items. A source-manifest script reconstructs the evaluated rows from the TIGER-Lab/MMLU-Pro test parquet snapshot and verifies saved identifiers against source category and answer labels. The manifest stores normalized item hashes and source identifiers, not benchmark question text. The reconstructed source revision is b189ec765aa7ed75c8acfea42df31fdae71f97be; the source parquet SHA256 is 0e24a191921c2f453518a537a8b2117bd137e7714d4ef1565e9ba06c1ecb9ad8; and the public manifest item-hash SHA256 is 1f482b880fad0d8c97aa08f83d250ffd43e715b78ec2619987e0db4e9a98d9d3.

All 3,008 evaluated identifiers match the source rows with zero category or answer mismatches. The subset is now reconstructable, but the original reason for selecting exactly these 3,008 rows from MMLU-Pro was not preserved. Because the subset-selection rationale was not preserved, no sampling inference is made from this subset. The subset is used only to exercise the raw-output audit pipeline.

The raw-output diagnostic illustrates what is required beyond a binary correctness matrix. Three runs include saved prediction fields; the remaining rows require deterministic extraction from raw responses. A conservative prompt-echo guard marks responses ending in a bare chat marker after echoed prompt text as unanswered. This

Field	Value
Evaluated items	3008
Matched MMLU-Pro rows	3008
Mismatches	0
Missing source rows	0
Manifest reconstruction passed	yes

Table 10. MMLU-Pro source-manifest reconstruction. The public artifact stores row hashes and source identifiers, not benchmark question text or answer-option text.

Component	Status	Main-claim use
Source reconstruction	3,008/3,008 item identifiers matched	Provenance support
Raw-response extraction	Deterministic parser plus scoring-mode sensitivity	Diagnostic only
Manual parser validation	Audit sample generated; labels incomplete	Not claimed
MMLU-Pro family comparison	Parser-dependent and incompletely manifested	Appendix diagnostic only

Table 11. Diagnostic status of the archived MMLU-Pro raw-output workflow. The WILD-supported core claim does not depend on MMLU-Pro parser validation.

Model	Saved acc.	Raw-only acc.	Δ	Saved answered	Raw answered	Changed corr.
Gemma-3-1B-Instruct	11.8	13.1	+1.4	99.1	99.0	11.6
Gemma-3-270M-Instruct	10.4	10.0	-0.4	95.5	95.5	3.3
Phi-2	11.8	10.9	-0.9	90.8	61.7	8.2

Table 12. Scoring-mode sensitivity for runs with both saved predictions and raw responses. Values are percentages. Δ is raw-response-only accuracy minus saved-prediction-primary accuracy.

matters for Gemma-4-E2B-Instruct: after the guard, its answered rate is 5.2% and parse-failure rate is 94.8%. That run is retained in descriptive files as a generation-format failure but excluded from primary pairwise claims.

The parser-audit sample has been generated but not labeled. Therefore raw-output parser validation is not claimed, and the MMLU-Pro results are not used as confirmatory evidence for model-family behavior. Table 11 summarizes the diagnostic status of the case study.

The scoring-mode sensitivity check uses the three runs that contain both saved predictions and raw responses. Table 12 compares the default saved-prediction-primary scoring with a raw-response-only parser. It shows that scoring asymmetry is materially important, not merely measurable: Gemma-3-1B-Instruct changes correctness on 11.6% of rows across the two modes, and Phi-2’s raw-response-only answerability is much lower than its saved-prediction-primary answerability. This sensitivity check is not a substitute for manual validation, but it prevents the paper from treating the parser as invisible.

The archived raw run files preserve item IDs, categories, ground-truth labels, model labels, and raw responses. They do not preserve exact Hugging Face repository identifiers, revision hashes, prompt templates, chat templates,

Source	Tier	Access	Paper use
WILD item-level correctness	B	public	core item-level evidence
WILD raw records	A	public_large	not redistributed
Open LLM Leaderboard aggregates	C	public	context only
Open LLM Leaderboard details	D	gated_terms_required	excluded unless access granted
OpenCompass predictions	D	gated_terms_required	excluded unless access granted
Model cards and reports	C	public_or_model-gated	context only

Table 13. External evidence map used by the artifact. Only Tier A/B item-level sources can support replacement-accounting metrics. Tier C sources are context only, and Tier D sources are logged as exclusions unless access is later granted.

decoding parameters, runtime versions, run dates, or complete hardware metadata. Consequently, the MMLU-Pro accounting can be regenerated from saved outputs, but the original inference runs cannot be independently reproduced as validated model-behavior evaluations without rerunning or reconstructing the experiment.

Compact MMLU-Pro extraction coverage, parser gates, run-metadata status, scoring sensitivity, and one same-family diagnostic summary are reported in the appendix. Full directed comparison tables, domain triage outputs, near-parity diagnostics, and generated figures are included in the public release rather than promoted in the manuscript PDF. They document artifact behavior and do not support confirmatory MMLU-Pro model-evaluation claims.

7 External Evidence Availability

The artifact classifies external evidence by what it can prove. This distinction is not bookkeeping; it is the boundary between valid replacement accounting and unsupported extrapolation. Error persistence, regression mass, and churn require paired item-level outcomes. Aggregate leaderboards can contextualize model quality, but they cannot identify corrected errors or newly introduced regressions.

Public model cards and technical reports provide useful aggregate context for small-model capability (Gemma Team 2024, 2025; Grattafiori et al. 2024; Qwen Team 2024; A. Yang et al. 2025), but they are not substitutes for paired item-level outputs. The artifact records source-backed aggregate scores and source pointers as context-only evidence in `analysis/external_benchmark_context/`. These scores may differ in prompt format, shot count, model variant, benchmark revision, and evaluation harness. For that reason, no paper claim about error persistence, regression mass, or churn is derived from external aggregate scores.

8 Discussion

8.1 What the evidence supports

The public WILD demonstration supports the core audit claim without answer extraction: aggregate gains can coincide with large item-level churn. The MMLU-Pro diagnostic is included to show how the artifact handles raw-response evidence with incomplete scoring validation. Family-level diagnostic summaries are provided only in the appendix and public release to document artifact behavior; they are not part of the article’s core claim set.

8.2 What the evidence does not support

The evidence does not support a universal scaling law, a mechanistic explanation of why families differ, or a claim that one family is generally better for deployment. It also does not support interface-tax, long-context, safety, tool-use, multimodal, or multi-turn claims. The WILD results are correctness-only. The MMLU-Pro results are parser-dependent until the manual audit is complete.

8.3 How practitioners should use the audit

For an actual deployment, the audit should be run on the proposed successor path and on the task distribution that matters for the product. A high-churn replacement may be acceptable when corrected items are more important than regressions. That is a product and risk decision, not a conclusion implied by aggregate accuracy.

The audit card should be treated as a triage tool. Large regression mass, high normalized regression burden, and near-parity churn should trigger manual inspection of affected item clusters before deployment.

9 Threats to Validity

Construct validity. The audit measures changes in binary correctness over shared labeled items. It does not directly measure usefulness, calibration, safety, open-ended quality, or user satisfaction. WILD avoids parser error but is limited to correctness records. MMLU-Pro uses raw outputs but depends on deterministic answer extraction.

Parser validity. The MMLU-Pro parser may misclassify some raw completions, especially fallback cases. The generated 452-row manual audit sample includes 120 high-risk rows, but it must still be human-labeled before the raw-output case study can be treated as a fully validated model-evaluation result. This is the primary blocker for treating the MMLU-Pro case study as validated model evaluation. Until then, MMLU-Pro churn and regression claims are correctness-under-parser claims. The saved-prediction sensitivity check shows that scoring mode can change per-row correctness for runs with both saved predictions and raw responses.

Internal validity. The archived MMLU-Pro run logs do not preserve exact repository identifiers, model revision hashes, prompt templates, chat templates, decoding parameters, runtime versions, run dates, or complete hardware metadata. The reported accounting can be regenerated from saved outputs, but model-behavior replication requires full run manifests or reruns.

External validity. The primary raw-output case study uses one fixed MMLU-Pro subset. WILD adds public cross-benchmark correctness evidence over a selected subset of models and tasks, but it does not include the same prompts, raw completions, or parser decisions. Additional raw-output datasets and model families are needed for stronger generalization.

Selection and direction. The MMLU-Pro all-pairs diagnostic uses a deterministic ordering rule, not a complete vendor release chronology. Directed comparisons marked as chronology unclear should not be interpreted as real deployment replacements.

10 Artifact Outputs and Reproduction

The artifact is organized around generated, inspectable files rather than hand-computed paper values. The MMLU-Pro pipeline emits pairwise metrics as CSV/JSON, replacement-audit cards as CSV/JSON/Markdown, quality-gate summaries, parser-audit summaries, scoring-mode sensitivity tables, source-manifest hashes, and paper-ready LaTeX tables. The WILD pipeline emits normalized correctness records, pairwise replacement metrics, task-family summaries, task-dispersion summaries, a source-coverage report, and paper-ready tables. The external-evidence pipeline emits a source-quality map, aggregate context tables, informal-context exclusions, and a claim-use manifest.

The artifact also includes a CPU-only mini-rerun harness for fully manifested follow-up validation. It writes a run manifest, environment file, selected-item hash manifest, resumable raw records, per-model summaries, pairwise audit metrics, and claim checks. A completed real-model mini-rerun is not used as evidence in the present manuscript unless its outputs, exact model revisions, prompt template, tokenizer/chat-template metadata, decoding settings, runtime environment, hardware summary, and run date are archived with the release.

Field	Value
Repository	https://github.com/DsChauhan08/error_persistence_regression_gap
Public source archive	https://archive.softwareheritage.org/browse/origin/?origin_url=https://github.com/DsChauhan08/error_persistence_regression_gap.git
Code license	MIT
Data-output license	CC-BY-4.0 for derived aggregate outputs; source datasets remain under their original terms
Python support	Python 3.11+ recommended; local run metadata is recorded in JSON
Dependency file	requirements.txt
Public tests	Passed
Public hygiene scan	Passed
Public manifest	Verified
WILD accounting layer	Supported by public item-level correctness data
MMLU-Pro source reconstruction	Supported by source-row hashes
Public proof-of-analysis package	Available
Methods/software article scope	Supported by the public artifact
MMLU-Pro parser validation	Not claimed in this article
MMLU-Pro confirmatory evaluation	Not claimed; diagnostic workflow only
Claim boundary	Parser-dependent MMLU-Pro model-evaluation claims are excluded from the article’s core claim set.

Table 14. Artifact availability and claim scope. Parser validation and MMLU-Pro confirmatory status remain visible but do not block the parser-independent WILD/source-manifest claim set for this software-tool article.

The release also includes tests for source-manifest matching, parser-audit summaries, scoring-mode sensitivity, WILD replacement accounting, evidence-tier restrictions, CPU-rerun resume behavior, and public-release hygiene. A hygiene scan checks that public artifacts do not contain raw model responses, benchmark text, local cache paths, or private files.

11 Data and Software Availability

The GitHub/SSRN public proof-of-analysis package is available at https://github.com/DsChauhan08/error_persistence_regression_gap. The package contains the manuscript, analysis code, derived MMLU-Pro pairwise metrics, source-row hashes, parser-audit metadata, scoring-mode sensitivity summaries, WILD correctness-only replication outputs, evidence-tier manifests, generated tables, and generated figures. It does not redistribute full MMLU-Pro question text, full answer-option text, raw model responses, or item-level response excerpts.

The accounting can be regenerated from saved outputs, but full model-behavior replication requires complete run manifests. A future rerun intended to support confirmatory model-evaluation claims should archive exact model repository IDs, revision hashes, prompt templates, chat templates, decoding parameters, inference engine, package versions, hardware summary, and run dates. The absence of completed manual parser labels is therefore not hidden or bypassed: parser-dependent MMLU-Pro claims are excluded from the core claim set in Table 2.

The main public-package reproduction commands are:

```
make reproduce-public
make test
make hygiene
make manifest
```

The `make reproduce-public` target regenerates the public WILD and evidence-registry outputs. The public package also exposes the underlying Python modules for users who want to rerun individual steps directly.

Parser-audit regeneration and scoring-mode sensitivity over private/source raw outputs require access to the withheld raw model-output JSONL files:

```
PYTHONPATH=src python -m boundary_slm.scoring_mode_sensitivity
PYTHONPATH=src python -m boundary_slm.parser_audit_impact
PYTHONPATH=src python -m boundary_slm.parser_audit_second_pass
PYTHONPATH=src python -m boundary_slm.mmlu_scoring_robustness
```

The public package keeps only redacted audit metadata and aggregate summaries.

12 Conclusion

This paper presents an item-level audit for small language model replacement. Aggregate accuracy says whether a candidate model wins on average. The proposed decomposition shows how it wins, what it fails to fix, and what it breaks.

The public WILD demonstration shows that the audit can be applied to independent item-level correctness data and that modest aggregate gains can still involve large churn. The MMLU-Pro archived-output diagnostic illustrates how the same audit workflow applies to raw-response evidence, while exposing the need for source manifests, parser validation, scoring-mode sensitivity, and full run provenance.

The practical recommendation is simple: benchmark-based model replacement decisions should report corrected errors, regressions, persistent errors, and churn mass, not only aggregate accuracy.

13 Author Contributions

Dhananjay Singh Chauhan designed the study, ran or assembled the model-output evaluations, implemented the analysis code, performed the statistical analysis, generated the tables and figures, and wrote the manuscript.

14 Competing Interests

The author declares that there are no competing interests.

15 Grant Information

The author declares that no external funding supported this work.

16 Acknowledgments

The author thanks Kaggle and Google Colab for making accessible compute environments available for the MMLU-Pro testing and related validation runs. The author also thanks the maintainers of the open-source model, benchmark, and evaluation ecosystems that made the raw model-output and public correctness-matrix analyses possible.

References

- E. Breck, S. Cai, E. Nielsen, M. Salib, and D. Sculley. 2017. “The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction.” In: *2017 IEEE International Conference on Big Data*, 1123–1132. doi:[10.1109/BigData.2017.8258038](https://doi.org/10.1109/BigData.2017.8258038).
- A. D’Amour et al.. 2022. “Underspecification Presents Challenges for Credibility in Modern Machine Learning.” *Journal of Machine Learning Research*, 23, 226, 1–61. <https://www.jmlr.org/papers/v23/20-1335.html>.
- T. G. Dietterich. 1998. “Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms.” *Neural Computation*, 10, 7, 1895–1923. doi:[10.1162/089976698300017197](https://doi.org/10.1162/089976698300017197).
- M. M. Fard, Q. Cormier, K. Canini, and M. Gupta. 2016. “Launch and Iterate: Reducing Prediction Churn.” In: *Advances in Neural Information Processing Systems*. https://proceedings.neurips.cc/paper_files/paper/2016/hash/dc5c768b5dc76a084531934b34601977-Abstract.html.

- T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford. 2021. “Datasheets for Datasets.” *Communications of the ACM*, 64, 12, 86–92. doi:[10.1145/3458723](https://doi.org/10.1145/3458723).
- Gemma Team. 2024. “Gemma 2: Improving Open Language Models at a Practical Size.” *arXiv preprint arXiv:2408.00118*. <https://arxiv.org/abs/2408.00118>.
- Gemma Team. 2025. “Gemma 3 Technical Report.” *arXiv preprint arXiv:2503.19786*. <https://arxiv.org/abs/2503.19786>.
- A. Grattafiori et al.. 2024. “The Llama 3 Herd of Models.” *arXiv preprint arXiv:2407.21783*. <https://arxiv.org/abs/2407.21783>.
- D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. 2021. “Measuring Massive Multitask Language Understanding.” In: *International Conference on Learning Representations*.
- J. Kirkpatrick et al.. 2017. “Overcoming Catastrophic Forgetting in Neural Networks.” *Proceedings of the National Academy of Sciences*, 114, 13, 3521–3526. doi:[10.1073/pnas.1611835114](https://doi.org/10.1073/pnas.1611835114).
- M. Krundick, A. Wiemerslage, S. Ebner, C. Lovering, and C. Tanner. 2026. “Cost-Efficient Estimation of General Abilities Across Benchmarks.” *arXiv preprint arXiv:2604.01418*. Retrieved June 19, 2026 from <https://arxiv.org/abs/2604.01418>.
- P. Liang et al.. 2022. “Holistic Evaluation of Language Models.” *arXiv preprint arXiv:2211.09110*. <https://arxiv.org/abs/2211.09110>.
- Q. McNemar. 1947. “Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages.” *Psychometrika*, 12, 2, 153–157. doi:[10.1007/BF02295996](https://doi.org/10.1007/BF02295996).
- M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru. 2019. “Model Cards for Model Reporting.” In: *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229. doi:[10.1145/3287560.3287596](https://doi.org/10.1145/3287560.3287596).
- Qwen Team. 2024. “Qwen2.5 Technical Report.” *arXiv preprint arXiv:2412.15115*. <https://arxiv.org/abs/2412.15115>.
- M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh. 2020. “Beyond Accuracy: Behavioral Testing of NLP Models with CheckList.” In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–4912. <https://aclanthology.org/2020.acl-main.442/>.
- D. Sculley et al.. 2015. “Hidden Technical Debt in Machine Learning Systems.” In: *Advances in Neural Information Processing Systems*, 2503–2511. <https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems>.
- A. Srivastava et al.. 2022. “Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models.” *arXiv preprint arXiv:2206.04615*. <https://arxiv.org/abs/2206.04615>.
- Y. Wang et al.. 2024. “MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark.” *arXiv preprint arXiv:2406.01574*. <https://arxiv.org/abs/2406.01574>.
- Y. Xie, Y.-A. Lai, Y. Xiong, Y. Zhang, and S. Soatto. 2021. “Regression Bugs Are In Your Model! Measuring, Reducing and Analyzing Regressions In NLP Model Updates.” In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 6589–6602. doi:[10.18653/v1/2021.acl-long.515](https://doi.org/10.18653/v1/2021.acl-long.515).
- A. Yang et al.. 2025. “Qwen3 Technical Report.” *arXiv preprint arXiv:2505.09388*. <https://arxiv.org/abs/2505.09388>.

A MMLU-Pro Diagnostic Details

Family	Models	Rows	Saved pred.	Explicit marker	Fallback	Unanswered	Saved-field models
Gemma	5	15040	5855	4169	1929	3087	2
Llama	1	3008	0	2440	513	55	0
Phi	1	3008	2730	0	0	278	1
Qwen	7	21056	0	15282	5686	88	0

Table 15. Answer-source coverage by family. Saved predictions are used when present; otherwise explicit answer markers and fallback letter rules are applied to raw responses.

Validation item	Value
Audit sample rows	452
High-risk audit rows	120
Completed human-labeled rows	0
Saved-prediction consistency rows	8585
High-confidence sensitivity families	4
Manual audit ready	no

Table 16. Parser-validation status for the archived MMLU-Pro raw-output diagnostic. Manual labels are required before reporting parser agreement as a validation result.

Dimension	Value	Rows	Agree	False corr.	False incorr.	False unans.
overall	pending labels	0	–	–	–	–
Claim gate	blocked	0	–	0	0	–

Table 17. Parser-audit impact gate. The current artifact has generated the sample but has not completed human labels, so MMLU-Pro remains a parser-dependent diagnostic.

Mode	Family	Pairs	Mean N	Δ	R	Churn	Persist.
default	gemma	10	3008	+0.5	11.8	24.1	85.9
default	qwen	21	3008	+17.2	8.9	35.0	65.8
exclude_low_confidence_fallback	gemma	10	1633	+22.8	6.8	36.4	65.5
exclude_low_confidence_fallback	qwen	21	1747	+23.8	7.6	39.1	54.7
exclude_saved_prediction_models	gemma	3	3008	-10.5	19.0	27.6	89.5
exclude_saved_prediction_models	qwen	21	3008	+17.2	8.9	35.0	65.8
explicit_marker_only	gemma	10	1633	+22.8	6.8	36.4	65.5
explicit_marker_only	qwen	21	1747	+23.8	7.6	39.1	54.7
raw_only	gemma	10	3008	+0.4	11.8	23.9	86.0
raw_only	qwen	21	3008	+17.2	8.9	35.0	65.8

Table 18. MMLU-Pro scoring-robustness modes over archived outputs. These are sensitivity analyses, not confirmatory validation while parser labels are incomplete.

B MMLU-Pro Model Runs and Quality Gate

Model label	Family	Params (B)	Accuracy	Answered	Primary role
Qwen3-4B-Instruct-2507	Qwen	4.00	48.2	100.0	paired
Qwen3.5-2B	Qwen	2.00	46.9	99.8	paired
Qwen2.5-3B-Instruct	Qwen	3.00	44.2	99.7	paired
Qwen3.5-0.8B	Qwen	0.80	33.0	99.6	paired
Gemma-2-2B-Instruct	Gemma	2.00	31.7	99.4	paired
Qwen3-0.6B	Qwen	0.60	28.0	100.0	paired
Llama-3.2-1B-Instruct	Llama	1.00	21.2	98.2	anchor
Gemma-3-1B-it	Gemma	1.00	17.5	98.1	paired
Qwen2.5-0.5B-Instruct	Qwen	0.50	16.5	99.3	paired
Qwen2-0.5B-Instruct	Qwen	0.50	13.0	98.7	paired
Phi-2	Phi	2.70	11.8	90.8	anchor
Gemma-3-1B-Instruct	Gemma	1.00	11.8	99.1	paired
Gemma-3-270M-Instruct	Gemma	0.27	10.4	95.5	paired
Gemma-4-E2B-Instruct	Gemma	2.00	1.8	5.2	excluded

Table 19. Model-level leaderboard and quality-gate status for the archived MMLU-Pro outputs. Accuracy and answered values are percentages over the same 3,008-item set.

C MMLU-Pro Run Metadata Status

Metadata field	Captured for all models?	Models with field	Total models	Complete	Missing	Note
Item id	yes	14	14	14	0	captured
Category	yes	14	14	14	0	captured
Ground-truth label	yes	14	14	14	0	captured
Raw response	yes	14	14	14	0	captured
Saved prediction	no	3	14	3	11	missing/incomplete
GPU worker id	no	2	14	2	12	missing/incomplete
HF repository	no	0	14	0	14	missing/incomplete
Model revision hash	no	0	14	0	14	missing/incomplete
Prompt template	no	0	14	0	14	missing/incomplete
Chat template	no	0	14	0	14	missing/incomplete
Decoding parameters	no	0	14	0	14	missing/incomplete
Runtime/package versions	no	0	14	0	14	missing/incomplete
Run date	no	0	14	0	14	missing/incomplete

Table 20. Run-metadata completeness in the archived MMLU-Pro raw-output files.

D MMLU-Pro Same-Family Diagnostic Summaries

Family	Models	Pairs	Old acc.	New acc.	Δ	I	R	Churn	Err. persist.	Corr. rate	NRB
Gemma	4	6	12.1	23.6	11.6	19.5	7.9	27.3	77.8	22.2	9.0
Qwen	7	21	24.3	41.4	17.2	26.1	8.9	35.0	65.8	34.2	12.8

Table 21. Primary MMLU-Pro same-family diagnostic accounting after excluding runs with answered rate below 80%. Values are percentages. I is improvement mass, R is regression mass, and NRB is normalized regression burden.

Full MMLU-Pro directed comparison tables, near-parity diagnostics, domain triage outputs, descriptive family checks, and generated figures are included in the public release under `analysis/paper_metrics/` and `paper/figures/`. They are not part of the core WILD-supported claim set.

E Answer-Extraction Rules

Answer extraction is deterministic. If a row contains a saved prediction field equal to one valid option letter, that field is used. Otherwise the parser searches the raw response in the following order and returns the last matching valid option letter:

- (1) A single-letter response, optionally wrapped in parentheses or followed by a period.
- (2) Explicit final-answer markers: `final answer`, `correct answer`, `correct option`, `correct choice`, `therefore`, `answer is`, and `option is`.
- (3) Tail option markers in the final 500 characters of the response, such as a standalone option letter followed by punctuation.
- (4) The final standalone option letter in the response, used only as a low-confidence fallback.

Before these rules are applied, a prompt-echo guard marks a response as unanswered if it ends in a bare chat marker after echoed prompt text and contains no explicit answer marker. The valid option alphabet is A–J, matching the saved ground-truth labels. Rows with no extracted valid option are counted as unanswered and incorrect.